

Noah Shahraz

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EDUCATION

California Polytechnic State University, San Luis Obispo

San Luis Obispo, CA

B.S. Computer Science (in progress), Minor in Mathematics (in progress)

Expected May 2030

TECHNICAL SKILLS

Languages: Python, TypeScript, R, MATLAB

ML / Data: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Matplotlib

Web / Backend: Flask, React, RAG, Ollama

Tools: Git, GitHub, Claude Code, Codex

Coursework: Harvard CS50; Stanford / DeepLearning.AI Machine Learning Specialization (2026)

PROJECTS

MesoNet Deepfake Detection (paper reproduction) | *Python, PyTorch, R*

2026

- Reproduced MesoNet (Afchar et al., WIFS 2018) from scratch in PyTorch, implementing Meso-4 and MesoInception-4 (28K parameters each) and training on FaceForensics++ across three seeds on Apple Silicon MPS
- Matched the paper's per-video protocol within noise: 95.0–95.5% accuracy and 0.984–0.995 AUC, landing on the authors' reported 95.3% Face2Face figure at the same compression level
- Stress-tested generalization by freezing the trained model and evaluating on two unseen datasets, documenting a consistent below-chance ranking inversion (AUC 0.41) that a 20.8M-parameter Xception baseline reproduced more severely
- Neutralized the inversion by retraining on the union of two manipulation types (AUC 0.41 → 0.50); validated findings with a video-clustered mixed-effects model in R over 67,200 frames and bootstrap confidence intervals

Hold'em Trainer | *TypeScript, React, Vite*

2026

- Built a browser-based poker training tool with preflop range drills, an exact equity calculator, postflop decision quizzes, and full no-limit play against six bot archetypes
- Wrote a dependency-free engine: a 5–7 card evaluator collapsing any hand to a 24-bit integer for exact comparison, run-out enumeration with Monte Carlo fallback offloaded to a web worker, and correct side-pot and minimum-raise handling
- Removed modulo bias from dealing via rejection sampling over `crypto.getRandomValues` with Fisher-Yates, and shipped an in-app chi-squared fairness test over up to 200,000 shuffles
- Covered the engine with 105 tests including brute-force evaluator verification and chip conservation across hundreds of bot-driven hands

Agora | *Python, Flask, RAG, Ollama*

2026

- Built a retrieval-augmented philosophy engine that answers questions in a chosen philosopher's voice and grounds every claim in a clickable citation to the retrieved source passage
- Chunked and embedded primary texts from Plato, Aristotle, Kant, and Nietzsche into a local vector store, running inference on a local Llama 3.1 8B model with no API cost and full offline operation
- Implemented a debate mode where two to four philosophers give cited opening positions and then rebut each other by name across a second round

EXPERIENCE & LEADERSHIP

Schoolhouse.world

Remote

SAT Math Class Instructor

Apr. 2025 – May 2025

- Led a class of 10+ students through a structured SAT math prep program, designing and delivering lessons across algebra, geometry, advanced math, and data analysis
- Tracked progress through practice assessments, diagnosed weak areas, and provided individualized support that produced measurable score improvement

Gunn High School, Focus on Success

Palo Alto, CA

Teaching Assistant

Aug. 2022 – Dec. 2022

- Supported 20+ students through coursework with individualized academic support, concept clarification, and problem-solving guidance
- Collaborated with instructors to evaluate assignments, deliver constructive feedback, and support daily classroom operations

Mountain View High School Varsity Basketball

Mountain View, CA

Team Captain

2022 – 2026

- Four-year varsity player and senior team captain; De Anza A-League All-League and Scholar Athlete (4.0 GPA)
- Also competed nationally with Oakland Soldiers (Nike EYBL) while carrying a full academic course load

Harvesting Sustainability – Mobile Harvesters

Palo Alto, CA

Co-Founder & Community Outreach Lead

June 2025 – Aug. 2025

- Co-led a student-founded nonprofit that reduces local food waste by harvesting surplus backyard produce for donation
- Drove 2,000+ lbs of fresh produce donated to community members in need; earned a feature in the *Los Altos Town Crier*
- Built community partnerships and ran outreach and education to grow volunteer participation and harvest scheduling

Equinox

Palo Alto, CA

Personal Trainer (NASM Certified)

Apr. 2026 – June 2026

- Designed and delivered individualized resistance, plyometric, and mobility programs for clients ranging from first-time exercisers to competitive athletes

Cooksies Lifeguard & Swim Academy

Palo Alto, CA

Lifeguard

May 2025 – Present

- Maintained constant surveillance of an aquatic facility, performed water rescues and first aid response, and handled water chemistry and routine inspections

Kung Fu Tea

Palo Alto, CA

Boba Barista

Sept. 2024 – Nov. 2024

- Prepared specialty drinks in a high-volume environment and managed opening and closing procedures